Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Neuro-Symbolic Schema-Aligned GraphRAG for Automated Building and Urban-Scale Digital Twins			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (X)**, Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *Advanced AI modeling/decision-making, Industrial use and innovative AI*			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Energy / Environment Mechanical Systems	Energy/Environmental Mechanical Systems Software	Energy / Environment Mechanical Systems	Energy/Environmental Control Equipment

Concept of Proposed Technology	This research advances a novel class of schema-aligned GraphRAG (Graph-based Retrieval Augmented Generation) systems, originally developed by RPI in partnership with U.S.-based smart building government and industry stakeholders [1, 2]. The primary objective of this proposal is to translate this core research into a commercial-grade solution in the building, data center and urban energy industries, achieving TRL 7 (System Prototype Demonstration in Operational Environment) through immediate pilot deployment by Korean SME partners in the Korean and U.S. markets. The project leverages the investigators' existing research in schema and graph-based interoperability [3, 4, 5, 6] and semantic AI [7, 8] to provide commercialization ready technologies, while extending the current state of the art during latter phases of the research and development partnership. Our central research innovation lies in bridging the fidelity gap that currently limits the application of generative AI in Cyber-Physical Systems (CPS). While Large Language Models (LLMs) excel at processing unstructured text, they frequently lack the precision required for engineering tasks, resulting in plausible but functionally invalid system descriptions. Our approach resolves this by integrating Neuro-Symbolic AI, connecting the probabilistic power of LLMs with the deterministic rigor of verified data structures and knowledge graphs [9]. By operating over structured engineering data rather than text alone, this system ensures that generated outputs—ranging from energy simulation input files to control logic—are physically meaningful, syntactically valid, and directly executable. To enable the automated generation of high-fidelity building models, our research introduces a dual-method framework. This framework, which forms the core of the Phase 1 commercial pilot, integrates embedding alignment and representational learning with SHACL-constrained structured generation. This unification introduces a semantically grounded modeling paradigm in which AI systems act as reasoning agents capable of understanding, generating, and improving building energy models. Current semantic models often fail to capture the tacit "engineering intuition" inherent in complex systems. To address this, our embedding-alignment methodology encodes populations of building system definitions into high-dimensional latent spaces. In this vector space, geometric, material, and system interdependencies are preserved as topological embeddings [10]. By jointly aligning textual descriptions, simulation metadata, and relational graph representations, the model learns a shared semantic manifold linking design intent to physical configuration. Critically, this embedding space enables similarity-based retrieval and analogy reasoning. The AI does not merely query a database; it perceives the topological similarity between engineered systems. This allows for cross-domain transfer of building typologies, providing an empirical basis for inferring missing system information in legacy buildings where data is sparse — a common barrier in building energy systems [11]. While embeddings provide the inferential power, the generation process must be strictly governed to ensure simulation viability. We employ SHACL (Shapes Constraint Language) [12] not merely as a validation tool, but as a "semantic scaffold" for the generative process. Leveraging the EnergyPlus JSON (epJSON [13]) schema as a domain ontology, the model produces graph outputs that adhere to strict physical and logical constraints (e.g., conservation of mass, hierarchical system connectivity). Unlike standard RAG pipelines that retrieve text chunks, our system constrains the AI's token generation to valid graph edges and nodes. If the AI predicts a component configuration that violates a SHACL shape (e.g., a boiler connected directly to a cooling tower), the generation is blocked or rectified in real-time, ensuring syntactic validity and physical coherence. During the initial phase of joint R&D, the technology will be deployed to automate the authoring of valid building energy performance simulation models. Our approach treats the creation of simulation models and semantic mappings as a code-generation problem, but one in which the code is a structured knowledge graph describing the building systems through industry standard input formats such as epJSON– an emerging next generation standard for building level energy performance

simulation models.

To achieve this, we will design and deploy a distributed architecture of SHACL-based data transformers addressing the targeted engineering use cases of our Korean SME partners, potentially targeting commercial applications with potential applications to commercial buildings, data centers and larger urban energy modeling. These components will be capable of ingesting heterogeneous commercial data describing the buildings to be simulated and transforming this data into valid epJSON simulation models.

A key innovation in this phase is the use of the knowledge graph to generate synthetic datasets for fine-tuning the AI models. The AI does not just use the rules; it helps write them. LLMs will be used to infer patterns from existing datasets to suggest new SHACL transformation rules, which are then verified against domain-specific constraints. The LLM effectively "writes the validation logic" that subsequently confines its own output. This creates a self-correcting neuro-symbolic loop, allowing the system to rapidly adapt to new building types or regulatory requirements such as NY Local Law 97 [14] or Korean Zero Energy Building (ZEB) standards [15] without manual re-development of the simulation model. This capability lowers the technical threshold for practitioners, allowing non-experts to generate regulatory-grade energy models via natural language prompts.

Following the successful commercial validation of the model generation engine, the second phase of the research will expand the scope from individual file generation to ecosystem-wide orchestration. This phase introduces the Federated Semantic Infrastructure, moving beyond schema-driven constraints derived from existing ontologies to data-driven discovery.

We will leverage knowledge graph publishing, management, and analysis frameworks [35], to orchestrate a distributed intelligence network. The role of this graph is two-fold: (a) to federate data from diverse Building Information Modeling (BIM), Geographic Information Systems (GIS) and Internet of Things (IoT) systems and databases into a single unified semantically enabled data model [16], and (b) to support downstream tasks such as anomaly detection and causal performance inference [17, 18].

This approach deploys AI agents within the knowledge capture framework that act as active curators. These agents monitor the data streams in real-time, performing reasoning tasks and creating supplemental data nodes that enrich the graph dynamically.

A major barrier to the adoption of semantic web technologies in industry is the high cost of constructing the initial graph [19, 20]. Phase 2 addresses this by automating the Semantic Extract, Transform, and Load (SETL) process. We will employ data annotation techniques, including SETLr [21], RML [22], and Semantic Data Dictionaries (SDD) [23], to support the automated elicitation of entities. Figure 1 illustrates the organization of these key semantic infrastructure components, while Figure 2 presents our planned workflow for the capabilities supported by machine learning, where a knowledge graph created from multiple sources serves as the basis for model training and fine-tuning. In this advanced workflow, the AI analyzes raw, heterogeneous data sources and proposes the appropriate SDD mappings automatically. It identifies entities of interest and maps them to the appropriate ontology such as Brick [24] or Haystack [25]. This allows the system to scale continuously as new data streams are added, transforming energy modeling from a static, manual procedure into a continuously learning, knowledge-driven analytical ecosystem.

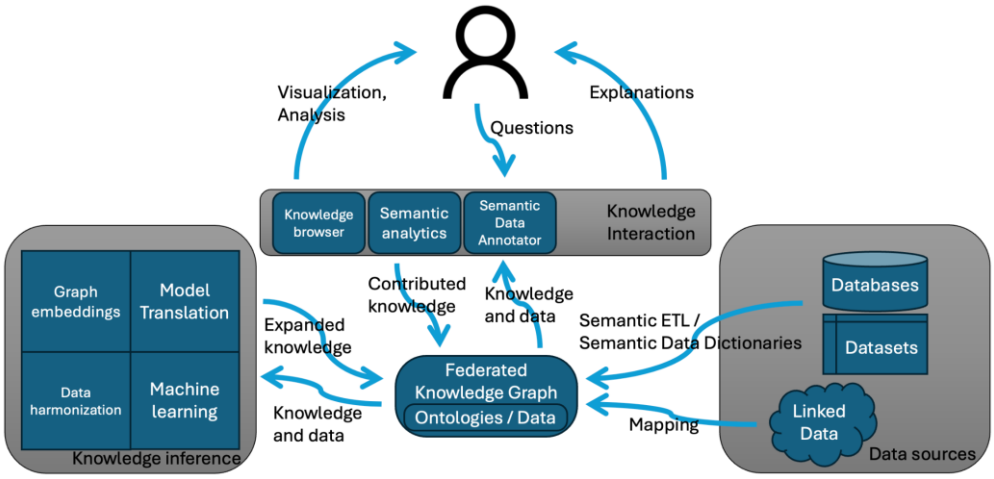

Figure 1 - Semantic Infrastructure

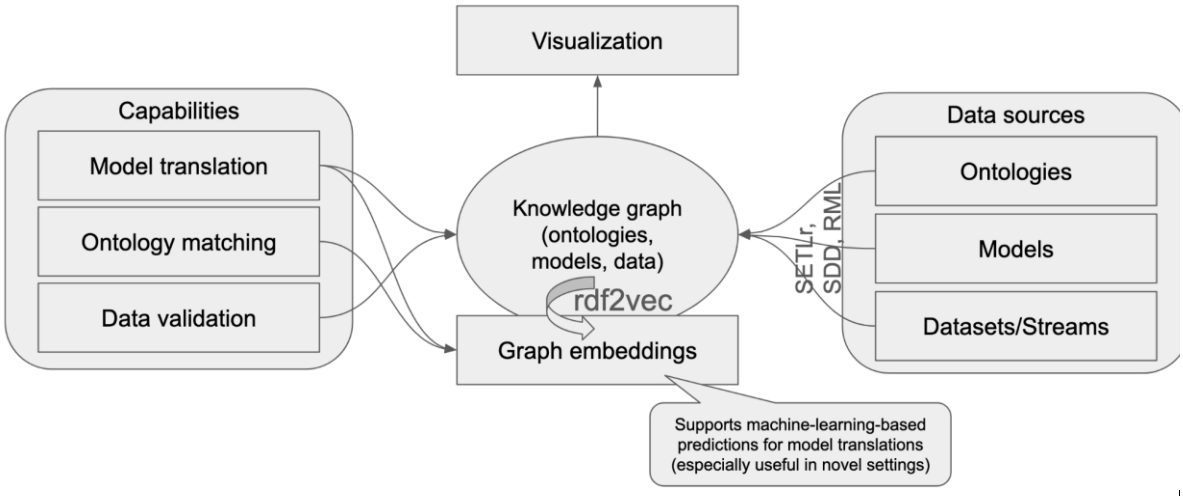

Figure 2 - Workflow of machine learning powered by knowledge graphs

In summary, this research advances the state of the art in the Korean building and urban energy industries – key industries where Korean SMEs as well as large multinational companies seek to establish leadership. In addition to these targeted industry advances, the neuro-symbolic schema-aligned GraphRAG approach represents significant advancement in AI modeling and decision making, with potential application broadly to industrial systems requiring data exchange and interoperability.

Rationale for Joint R&D Project	Korean SMEs are at the forefront of building energy management and controls systems, energy analytics, and smart city platforms, driven by regulatory requirements such as Zero Energy Building (ZEB) certification and expanding portfolio-scale compliance mandates. To a significant extent, these Korean industries are founded on energy modeling systems developed in the U.S. and informed by an ecosystem of international data standards and energy performance guidelines. The engineering model descriptions are difficult, time consuming and error prone for human engineers to develop. AI techniques show promise for automating much of this work, however current AI tools lack the reliability required to automate model generation, baseline definition, and control integration across heterogeneous building portfolios. This joint R&D project enables co-development of a commercialization-ready GraphRAG infrastructure that directly addresses these gaps. Collaboration is essential to validate the approach against Korean regulatory contexts, deploy pilots within SME platforms, and translate research outputs into exportable AI-enabled products.
Key Words	1. GraphRAG 2. Schema-Constrained AI 3. Automated Model Generation 4. Federated Energy & IoT Data 5. Control-Ready Digital Twins

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (4)	**End of the Project**	Phase 1: Level (6) Phase 2: Level (7)
R&D Objectives	• Develop a schema-aligned GraphRAG pipeline for generation of JSON energy models. • Establish Levels of Detail (LoD) suitable for compliance, portfolio analytics, and control use cases. • Demonstrate automated baseline generation, simulation execution, and reporting. • Enable control-ready simulation environments for MPC/RL evaluation.			
R&D Target Specifications	• GraphRAG Feasibility: Demonstrate that schema-aligned GraphRAG can reliably constrain AI generation over structured building energy and IoT data compared to text-only approaches. • Model Executability: Show repeatable generation of simplified, executable EnergyPlus epJSON models within a defined LoD-B scope, suitable for baseline simulation and scenario comparison. • Level of Detail Definition: Formally define and consistently apply a compliance-grade (LoD-B) modeling specification, including documented assumptions and boundary conditions. • Baseline Reasonableness: Demonstrate that AI-generated models produce directionally reasonable baseline energy results appropriate for screening, certification pre-checks, and planning use cases. • Traceability: Enable end-to-end traceability linking prompts, assumptions, models, simulation outputs, and reports through a shared knowledge graph. • Calibration Feasibility: Demonstrate exploratory calibration feasibility using standard metrics where data is available, without requiring full audit-grade tuning.			

	• Scalability: Validate the end-to-end pipeline across multiple representative buildings or archetypes and document scaling considerations for Phase 2, including the ability for large scale urban energy model generation.
R&D Plan	The Phase 1 research and commercialization plan is organized as follows. 1. Schema Alignment & Knowledge Graph Foundation (Months 1–2) Objective: Establish the structured foundation for GraphRAG and define LoD-B scope. • RPI: Lead schema alignment, ontology definition, and LoD-B specification; design initial knowledge graph structure. • Korean SME: Provide representative data schemas, regulatory context (e.g., ZEB/BEMS), and target platform requirements; review LoD relevance. Milestone: Validated LoD-B definition and initial schema-aligned knowledge graph. 2. GraphRAG Pipeline Development (Months 2–3) Objective: Demonstrate schema-constrained AI reasoning o`ver structured engineering data. • RPI: Develop GraphRAG pipelines, retrieval strategies, and schema-constrained prompt structures. • Korean SME: Provide use-case prompts and review output relevance for commercial workflows. Milestone: Evidence that GraphRAG improves consistency and validity relative to text-only approaches. 3. Automated epJSON Generation & Execution (Months 3–4) Objective: Validate automated generation of executable energy models. • RPI: Implement JSON model generation workflows and integrate simulation execution. • Korean SME: Validate applicability to compliance, baseline, and planning use cases. Milestone: Multiple AI-generated epJSON models successfully executed in EnergyPlus. 4. Baseline Simulation, Calibration Feasibility & Traceability (Months 4–5) Objective: Demonstrate baseline usefulness and explainability. • RPI: Implement traceability across prompt → graph → model → simulation → report. • Korean SME: Review outputs for certification, reporting, and platform integration relevance. Milestone: Demonstrated baseline reasonableness and end-to-end traceability. 5. POC Validation (Months 5–6) Objective: Validate feasibility across cases and define commercialization pathway. • RPI: Synthesize technical findings and define Phase 2 roadmap (LoD-C, control, scale). • U.S. Research Partner(s): Contribute evaluation results and risk assessment. • Korean SME: Define productization requirements, integration strategy, and go-to-market priorities. Milestone: validation report and jointly defined further development plan.

4.Role & Responsibility (R&R)

Expected R&R of Research Institute	• Lead GraphRAG architecture, schema alignment, and AI validation methods. • Define LoD specifications, KPIs, and evaluation protocols. • Provide existing expertise in federated building energy modeling and controls data standards and integration, open data protocols and data architecture. • Provide reference implementations and pilot demonstrations.
Expected R&R of Korean SME Partner	• Provide operational datasets, regulatory context, and pilot deployment environments. • Integrate outputs into commercial BEMS, analytics, or smart city platforms. • Lead productization, and customer validation.

5. Expected Outcome/Impact

Expected Outcomes and Impact	The project will deliver a commercialization-ready GraphRAG infrastructure that enables reliable AI reasoning over building energy and IoT systems. Expected impacts include reduced modeling and certification costs, accelerated deployment of BEMS and compliance solutions, and new AI-driven products for data centers, building portfolios, and smart cities. The collaboration will generate joint IP and strengthen Korean SME competitiveness in global AI and energy technology markets.

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☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization